

Python Mastery: The Core Foundations

Comprehensive Reference & Hands-on Examples

1. Python Syntax & Indentation

Python uses indentation to indicate a block of code. Missing indentation will cause an error.

```
age = 20
if age >= 18:
    print("Access Granted") # Indented by 4 spaces
    if age >= 21:
        print("Premium Access") # Nested indentation
```

2. Comments

Comments start with a # and are ignored by Python. Triple quotes are used for multi-line documentation.

```
# This is a single-line comment
"""
This is a multiline comment
explaining the logic below.
"""
```

3. Variables & Casting

Variables are containers for storing data. Casting is used to specify or change a variable's type.

```
# Handling price input as a float
item_price = "19.99"
tax_rate = 0.05
total = float(item_price) * (1 + tax_rate)
print(f"Total price: ${total:.2f}")
```

4. Operators

Category	Operator	Meaning
Arithmetic	//	Floor Division (returns integer)
Arithmetic	%	Modulus (returns remainder)
Arithmetic	**	Exponentiation (Power)
Logical	and, or, not	Combine conditional statements

5. Python Collections

List [Ordered, Changeable]

```
guests = ["Alice", "Bob"]; guests.append("Charlie")
```

Dictionary {Key:Value}

```
user = {"name": "Alice", "role": "Admin"}; print(user["role"])
```

6. Conditional Logic

```
score = 85
if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
else:
    grade = "C"

# Ternary Operator (Shorthand)
status = "Passing" if score >= 70 else "Failing"
```

Practical Exercise: Library Inventory

```
inventory = {"1984": 3, "Hamlet": 0}
request = "1984"

if request in inventory and inventory[request] > 0:
    print("Available!")
    inventory[request] -= 1
else:
    print("Out of stock")
```